

1. SETS

Definition:

A **set** is a collection of objects, called **elements** of the set. For example:

1. $A = \{\text{Nicosia, Larnaca, Limassol}\}$
2. $B = \{2, 4, 7, 11\}$

- Elements are contained in **braces**
- Sets are denoted by a capital letter for reference purposes. For example: $B = \{7, 2, 11, 4\}$.

Note: The set B is the same as the one listed above. **The order of the elements is irrelevant.**

- A set is called **finite** if it has finitely many elements.
- A set is called **infinite** if it has infinitely many elements.

Notation

- If x is an element of the set A , we write: $x \in A$ which reads: “The element x *belongs* to the set A ”.
- If x is NOT an element of the set A , we write: $x \notin A$ which reads: “The element x *does not belong* to the set A ”.

We can describe a set using one of the following methods:

(a) A **listing** or **enumeration** of the elements

For example:

$$A = \{1, 3, 5, 7, \dots\}$$

(b) **Predicate Form** or **Set Builder Notation**.

For example:

$$A = \{x: x \text{ is a positive odd integer}\}$$

Commonly Used Sets:

Certain sets of numbers occur very frequently that are denoted by a particular name and symbol:

U is the **universal set**.

$\emptyset$ or $\{\}$ is the **empty set**.

$\mathbb{N} = \{1, 2, 3, 4, \dots\}$ is the set of all **natural numbers**.

$\mathbb{N}_0 = \{0, 1, 2, 3, 4, \dots\}$ is the set of all **natural numbers** including zero.

$\mathbb{Z} = \{0, \pm 1, \pm 2, \pm 3, \dots\}$ is the set of all **integers**.

$\mathbb{Q} = \left\{\frac{p}{q}: p, q \in \mathbb{Z}, q \neq 0\right\}$ is the set of all **rational numbers**.

$\mathbb{R}$ is the set of all **real numbers**.

Example 1:

Enumerate the elements of the following sets:

(a) $A = \{x: x \in \mathbb{Z}, x^2 + 4x = 12\}$

(b) $B = \{n^2: n \in \mathbb{Z}\}$

(c) $C = \{x: x \in \mathbb{N}, 2 \leq x^2 \leq 26\}$

(d) $D = \{x: x \in \mathbb{Z}, 2 \leq x^2 \leq 26\}$

Solution:

(a) $x^2 + 4x = 12 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x - 2)(x + 6) = 0$
 $\Rightarrow x = 2 \text{ or } x = -6$

Hence, $A = \{-6, 2\}$.

(b) $B = \{0, 1, 4, 9, 16, 25, \dots\}$

(c) $C = \{2, 3, 4, 5\}$

(d) $D = \{-5, -4, -3, -2, 2, 3, 4, 5\}$

Example 2:

Express each of the following sets in predicate form:

(a) $A = \{2, 5, 8, 11, \dots\}$

(b) $B = \left\{1, \frac{1}{4}, \frac{1}{9}, \frac{1}{16}, \dots\right\}$

Solution:

(a) $A = \{3n - 1: n \in \mathbb{N}\}$

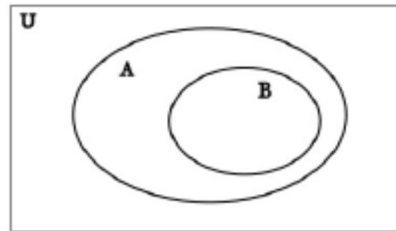
(b) $B = \left\{\frac{1}{n^2}: n \in \mathbb{N}\right\}$

Definition:

A set B is a **subset** of a set A if every element of B is an element of A . This is denoted by

$$B \subseteq A$$

A can be shown pictorially by means of a **Venn Diagram**.



For example, if $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 4\}$, then:

$$B \subseteq A \subseteq \mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

Definition:

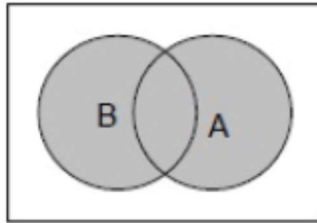
Two sets are equal if each is a subset of the other. In other words, two sets are equal if they contain exactly the same elements.

We write: $A = B$.

Set Operations

1. The **union** of two sets A and B is the set

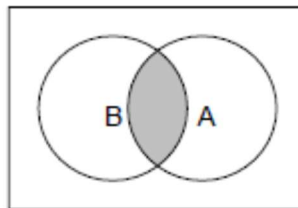
$$A \cup B = \{x: x \in A \text{ or } x \in B\}$$



It consists of the elements that belong to A , or to B , or both A and B .

2. The **intersection** of two sets A and B is the set

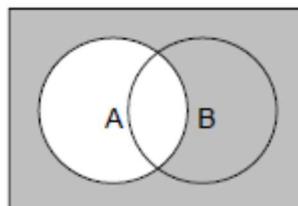
$$A \cap B = \{x: x \in A \text{ and } x \in B\}$$



It consists of the elements that belong to both sets A and B .

3. The **complement** of a set A is the set

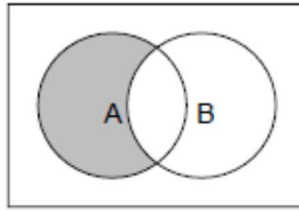
$$\bar{A} = A^c = A' = \{x: x \notin A\}$$



It consists of all the elements that do not belong to A .

4. The **complement of a set A relative to a set B** is the set

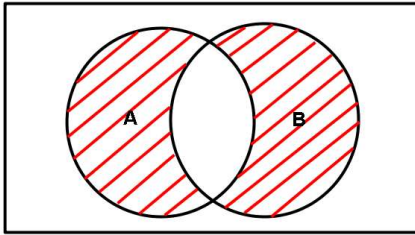
$$A - B = A \cap \bar{B} = \{x: x \in A, x \notin B\}$$



It consists of all the elements that belong A but do not belong to B .

5. The **symmetric difference** of two sets A and B is the set

$$A \triangle B = \{x: x \in A \text{ or } x \in B, x \notin A \cap B\}$$



It consists of the elements that belong to A or B but do not belong to both A and B .

Note:

$$A \triangle B = (A - B) \cup (B - A)$$

Example 3:

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$, $A = \{1, 3, 5, 7, 9\}$,
 $B = \{2, 4, 6, 8, 10\}$ and $C = \{3, 6, 9, 12\}$.

Find (a) $A \cup C$, (b) $B \cap C$, (c) $A - C$, (d) $\bar{A}$, (e) $B \triangle C$.

Solution:

(a) $A \cup C = \{1, 3, 5, 6, 7, 9, 12\}$

(b) $B \cap C = \{6\}$

(c) $A - C = \{1, 5, 7\}$

(d) $\bar{A} = \{2, 4, 6, 8, 10, 11, 12\}$

(e) $B \triangle C = \{2, 3, 4, 8, 9, 10, 12\}$

Example 4:

Let the universal set $U = \{n: n \in \mathbb{N}, 1 \leq n \leq 10\}$. Let

$$A = \{n: n \in \mathbb{N}, 2 \leq n \leq 6\}$$

$$B = \{n: n \in \mathbb{N}, 4 \leq n \leq 9\}$$

$$C = \{2n: n \in \mathbb{N}, 1 \leq n \leq 5\}$$

Find (a) $A \cup C$, (b) B^c , (c) $B - A$, (d) $A \cap B \cap C$, (e) $(A \cup B) \cap C$, (f) $(A \cap B)^c$, (g) $(A \cap B) \cup (B \cap C)$, (h) $A \triangle B$ and (i) $(A \triangle B) \cap C^c$

Solution:

$$(a) A \cup C = \{2, 3, 4, 5, 6, 8, 9, 10\}$$

$$(b) B^c = \{1, 2, 3, 10\}$$

$$(c) B - A = \{7, 8, 9\}$$

$$(d) A \cap B \cap C = \{4, 6\}$$

$$(e) (A \cup B) \cap C = \{2, 3, 4, 5, 6, 7, 8, 9\} \cap \{2, 4, 6, 8, 10\} = \{2, 4, 6, 8\}$$

$$(f) (A \cap B)^c = \{4, 5, 6\}^c = \{1, 2, 3, 7, 8, 9, 10\}$$

$$(g) (A \cap B) \cup (B \cap C) = \{4, 5, 6\} \cup \{4, 6, 8\} = \{4, 5, 6, 8\}$$

$$(h) A \triangle B = (A - B) \cup (B - A) = \{2, 3\} \cup \{7, 8, 9\} = \{2, 3, 7, 8, 9\}$$

$$(i) (A \triangle B) \cap C^c = \{2, 3, 7, 8, 9\} \cap \{1, 3, 5, 7, 9\} = \{3, 7, 9\}$$

Example 5:

Consider the following subsets of $\mathbb{N}$.

$$A = \{2n - 1 : n \in \mathbb{N}\}, \quad B = \{3n : n \in \mathbb{N}\}, \quad C = \{n : n \in \mathbb{N}, 10 \leq n \leq 20\}$$

Express each of the following in terms of A, B and C .

- (a) {the set of all even natural numbers}
- (b) {the set of all odd multiples of 3}
- (c) {11, 13, 15, 17, 19}
- (d) {10, 12, 14, 16, 18, 20}
- (e) {10, 14, 16, 20}
- (f) {15}
- (g) {12, 18}

Solution:

- (a) A^c
- (b) $A \cap B$
- (c) $A \cap C$
- (d) $A^c \cap C$
- (e) $C \cap (A \cup B)^c$
- (f) $A \cap B \cap C$
- (g) $(B \cap C) \cap A^c$

The Laws of Algebra of Sets

1. Commutative Laws

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

2. Associative Laws

$$A \cup (B \cup C) = A \cup B \cup C = (A \cup B) \cup C$$

$$A \cap (B \cap C) = A \cap B \cap C = (A \cap B) \cap C$$

3. Distributive Laws

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

4. Identity Laws

$$A \cup \emptyset = A$$

$$A \cup U = U$$

$$A \cap U = A$$

$$A \cap \emptyset = \emptyset$$

5. Idempotent Laws

$$A \cup A = A$$

$$A \cap A = A$$

6. Complement Laws

$$A \cup A^c = U \quad U^c = \emptyset \quad (A^c)^c = A$$

$$A \cap A^c = \emptyset \quad \emptyset^c = U$$

7. De Morgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

Example 6:

Use the laws of the algebra of sets to prove that for any sets A and B :

$$(a) \quad (A \cap B^c)^c \cup B = A^c \cup B$$

$$(b) \quad (A \triangle B) = (A \cup B) \cap (A \cap B)^c$$

Solution:

$$\begin{aligned}
 (a) \quad \text{LHS} &= (A \cap B^c)^c \cup B \\
 &= [A^c \cup (B^c)^c] \cup B && [\text{De Morgan's Laws}] \\
 &= (A^c \cup B) \cup B && [\text{Complement Laws}] \\
 &= A^c \cup (B \cup B) && [\text{Associative Laws}] \\
 &= A^c \cup B && [\text{Idempotent Laws}]
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \text{RHS} &= (A \cup B) \cap (A \cap B)^c \\
 &= (A \cup B) \cap (A^c \cup B^c) && [\text{De Morgan's Laws}] \\
 &= [(A \cup B) \cap A^c] \cup [(A \cup B) \cap B^c] && [\text{Distributive Laws}] \\
 &= [A^c \cap (A \cup B)] \cup [B^c \cap (A \cup B)] && [\text{Commutative Laws}] \\
 &= [(A^c \cap A) \cup (A^c \cap B)] \cup [(B^c \cap A) \cup (B^c \cap B)] && [\text{Distributive Laws}] \\
 &= [\emptyset \cup (A^c \cap B)] \cup [(B^c \cap A) \cup \emptyset] && [\text{Complement Laws}] \\
 &= (A^c \cap B) \cup (B^c \cap A) && [\text{Identity Laws}] \\
 &= (B - A) \cup (A - B) && [\text{Commutative Laws}] \\
 &= A \triangle B
 \end{aligned}$$

Definition:

The **cardinality** of a finite set A is the number of elements in A , and is denoted by $|A|$

Principle of inclusion and exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 7:

Each of the 63 first-year students can study a number of optional units. If 16 students chose to study Accounting, 37 chose to study Business and 5 chose to study both Accounting and Business, how many students took neither Accounting nor Business?

Solution:

Let $A = \{\text{accounting}\}$ and $B = \{\text{business}\}$. Then $A \cap B = \{\text{both}\}$.

Hence $|A| = 16$, $|B| = 37$ and $|A \cap B| = 5$. Then:

$$|A \cup B| = |A| + |B| - |A \cap B| = 16 + 37 - 5 = 48$$

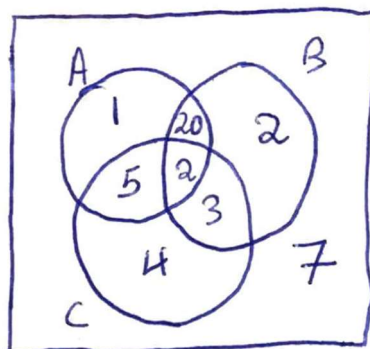
Therefore $63 - 48 = 15$ students took neither Accounting nor Business.

Problems similar to the above can also be solved with the aid of a Venn Diagram.

Example 8:

Each of the 44 first-year students can study a number of optional units. Suppose that 28 chose to study Accounting, 27 chose to study Biology and 14 chose to study Chemistry. Suppose that 22 students studied both Accounting and Biology, 7 studied both Accounting and Chemistry and 5 studied both Biology and Chemistry. 2 students studied all three options.

- (a) How many students studied at least one of the three options?
- (b) How many students studied none of the three options?
- (c) How many students only studied Chemistry?
- (d) How many students studied Biology but not Accounting?
- (e) How many students studied Chemistry or Biology but not Accounting?

Solution:

(a) $|A \cup B \cup C| = 37$

(b) $|(A \cup B \cup C)^c| = 44 - 37 = 7$

(c) $|C \cap (A \cup B)^c| = 4$

(d) $|B - A| = 5$

(e) $|(B \cup C) \cap A^c| = 9$

Definition:

An **ordered pair** is an element of the form (a, b) where a is an element of some set A and b is an element of some set B . The set of all ordered pairs of this form is called the **Cartesian product** of the two sets A and B and is denoted by $A \times B$. Hence,

$$A \times B = \{(a, b): a \in A \text{ and } b \in B\}$$

This operation leads to the concepts of relations and functions that we will see in our next two chapters.

Note: The cardinality of the Cartesian product of two sets A and B equals the product of the cardinalities of the two sets.

$$|A \times B| = |A| \times |B|$$

Example 9:

Let $A = \{x, y\}$ and $B = \{1, 2, 3\}$. Find the Cartesian products $A \times B$, $B \times A$ and $B \times B$ and state the cardinality of each.

Solution:

$$A \times B = \{(x, 1), (x, 2), (x, 3), (y, 1), (y, 2), (y, 3)\}$$

$$|A \times B| = |A| \times |B| = 2 \times 3 = 6$$

$$B \times A = \{(1, x), (2, x), (3, x), (1, y), (2, y), (3, y)\}$$

$$|B \times A| = |B| \times |A| = 3 \times 2 = 6$$

Note: $A \times B \neq B \times A$

$$B \times B = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

$$|B \times B| = |B| \times |B| = 3 \times 3 = 9$$

The Cartesian product of any collection of sets $A_1, A_2, \dots, A_n$ is the set

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i, i = 1, 2, \dots, n\}$$

The elements of these sets are finite lists and are the objects manipulated in many programming languages.

If $A_1, A_2, \dots, A_n$ are all the same set A , then we write A^n for the Cartesian product of n copies of A . For example, if $A = \{0,1\}$, then the elements of A^n are lists of zeros and ones of length n . These are called **bit strings** of length n .

Bit strings can be used to model set operations of finite sets. Let $S = \{s_1, s_2, \dots, s_n\}$ and $A \subseteq S$. We associate with A an n -bit string $(b_1, b_2, \dots, b_n)$ where $b_i = 1$ if $s_i \in A$ and $b_i = 0$ if $s_i \notin A$. This bit string is called the **characteristic vector** of A . Set operations can now be simulated by performing logical operations on the components of the corresponding characteristic vectors, provided we interpret 1 as “True” and 0 as “False”.

It is easier to perform the set operations first and then find the associated characteristic vector, so at this point we will only be interested in finding characteristic vectors of sets.

Example 10

Let $S = \{1, 2, 3, 4, 5\}$, $A = \{1, 3, 5\}$ and $B = \{3, 4\}$. Find the characteristic vectors of A and B , and determine the characteristic vectors of $A \cup B$, $A \cap B$, B^c and $A \cap B^c$.

Solution:

The characteristic vectors of A and B are

$$a = (1, 0, 1, 0, 1), \quad b = (0, 0, 1, 1, 0)$$

respectively.

The characteristic vector of $A \cup B = \{1, 3, 4, 5\}$ is:

$$a \text{ or } b = (1, 0, 1, 0, 1) \text{ or } (0, 0, 1, 1, 0) = (1, 0, 1, 1, 1)$$

The characteristic vector of $A \cap B = \{3\}$ is:

$$a \text{ and } b = (1, 0, 1, 0, 1) \text{ and } (0, 0, 1, 1, 0) = (0, 0, 1, 0, 0)$$

The characteristic vector of $B^c = \{1, 2, 5\}$ is:

$$\text{not } b = \text{not } (0, 0, 1, 1, 0) = (1, 1, 0, 0, 1)$$

The characteristic vector of $A \cap B^c = \{1, 5\}$ is:

$$a \text{ and } (\text{not } b) = (1, 0, 1, 0, 1) \text{ and } (1, 1, 0, 0, 1) = (1, 0, 0, 0, 1)$$